



Dynamics of intramammary infections and their association with quarter-level somatic cell score, milk yield, and milk components. Part 2: Non-*Staphylococcus* microorganisms

Mariana Fonseca,^{1,2}  Daryna Kurban,^{1,2}  Jean-Philippe Roy,^{2,3}  and Simon Dufour^{1,2*} 

¹Department of Pathology and Microbiology, Faculty of Veterinary Medicine, Université de Montréal, Saint-Hyacinthe, QC, Canada J2S 2M2

²Regroupement FRQNT Op+lait, Saint-Hyacinthe, QC, Canada J2S 2M2

³Department of Clinical Sciences, Faculty of Veterinary Medicine, Université de Montréal, Saint-Hyacinthe, QC, Canada J2S 2M2

ABSTRACT

Intramammary infections are often associated with elevated SCC, a key indicator of udder inflammation and milk quality. This condition leads to substantial economic losses, primarily through its direct impact on milk production. In addition to reducing milk yield, IMI can negatively affect milk components such as fat, protein, and lactose. Consequently, the main objective of this longitudinal study was to estimate the association of IMI caused by these microorganisms on the SCS, milk yield (kg/d), and milk components at the quarter level. A secondary objective was to describe the quarter-level incidence and elimination rates of IMI caused by microorganisms other than *Staphylococcus* spp. A convenience sample of 5 dairy herds using an automated milking system was selected and visited every 2 wk for milk sample collection. Daily quarter milk yield data were extracted from the automatic milking system software. The determination of SCC and milk composition parameters was performed by Lactanet. Bacteriological culture was performed according to the National Mastitis Council guidelines. The different types of colonies were counted and identified using MALDI-TOF. For the first objective, we used generalized linear models with robust variance to estimate the incidence rate and elimination rate of IMI. For the second objective, given the hierarchical structure of the data, a 4-level linear mixed model with herd, cow, and quarter as random intercepts was built with either the test-day SCS, milk yield, lactose, protein, or fat as the outcome. The most prevalent non-*Staphylococcus* microorganisms in primiparous cows were *Aerococcus viridans*, *Streptococcus dysgalactiae*, and *Corynebacterium bovis* with mean prevalences (95% CI) of 3.8% (3.3%, 4.4%), 0.9% (0.7%, 1.3%), and 0.9% (0.7%,

1.2%), respectively. Similarly, for multiparous cows, the 3 most prevalent pathogens were *C. bovis*, *A. viridans*, and *S. dysgalactiae*, with mean prevalences (95% CI) of 5.2% (4.8%, 5.5%), 3.7% (3.4%, 4.0%), and 2.8% (2.5%, 3.1%), respectively. For both primiparous and multiparous cows, the incidence rate of *S. dysgalactiae* was low, at 0.1 and 0.2 new intramammary infections/100 quarter-weeks, respectively. However, quarters infected with *S. dysgalactiae* exhibited a lower elimination rate compared with infections caused by other microorganisms. A similar pattern was observed for *S. uberis* infections in multiparous cows. In primiparous cows, quarters infected by *S. dysgalactiae* produced 1.3 kg/d less compared with healthy quarters. In multiparous cows, milk yield was not significantly associated with IMI status; however, quarters infected with *S. dysgalactiae* produced 0.7 kg/d less than healthy quarters, though this difference was only numerical. Regarding milk composition, quarters infected by *S. dysgalactiae* in primiparous cows had a 0.2% reduction in lactose content, with no significant differences observed for protein or fat content. In multiparous cows, *S. uberis* infections resulted in a 1.0% reduction in fat content, whereas *S. dysgalactiae* infections led to a 0.2% reduction in lactose content. Intramammary infections caused by *S. dysgalactiae* had a moderate impact on milk yield, particularly in primiparous cows. Although the effects on milk composition were minimal, the reductions in milk production underscore the importance of managing infections to mitigate economic losses in dairy herds.

Key words: dairy cattle, mastitis, Canada, milk production

INTRODUCTION

Mastitis is the most prevalent disease affecting dairy cows in North America, resulting in substantial economic losses for the dairy industry due to its direct impact on milk production and the costs associated with treatment

Received October 14, 2025.

Accepted March 18, 2026.

*Corresponding author: simon.dufour@umontreal.ca

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

(Ruegg, 2022a). Beyond the reduction in milk yield, mastitis adversely affects key milk components, including fat, protein, and lactose (Costa et al., 2019; Franzoi et al., 2020). Given that milk production is the primary income source for dairy farms, effective mastitis management is essential. In Canada, mastitis results in substantial economic losses, with an estimated annual cost of Can\$662 (US\$486) per dairy cow (Aghamohammadi et al., 2018). In addition to economic repercussions, mastitis also compromises the health and welfare of dairy cows (von Keyserlingk et al., 2009).

Understanding the epidemiology of mastitis is crucial for its prevention and control. It is well described that staphylococci and streptococci are among the most prevalent microorganisms associated with IMI worldwide (Piepers et al., 2007; Vakkamäki et al., 2017; Kurban et al., 2022). In Canada, *Streptococcus dysgalactiae* and *Streptococcus uberis* were among those with the highest incidence of clinical mastitis in 2005 (Olde Riekerink et al., 2008). Moreover, recent studies indicate that *S. dysgalactiae* and *S. uberis* are not only prevalent in clinical mastitis, but are also frequently isolated from apparently healthy mammary quarters. In a recent mapping review encompassing culture results from both the milk of apparently healthy mammary quarters and quarters having clinical mastitis of dairy cows worldwide, *S. dysgalactiae* and *S. uberis* were among the most frequently isolated microorganisms (Kurban et al., 2022). Other microorganisms also frequently identified in samples from apparently healthy mammary quarters in this same study were *Aerococcus viridans*, *Corynebacterium bovis*, *Corynebacterium amycolatum*, *Corynebacterium xerosis*, and *Lactococcus garvieae*.

Intramammary infections are commonly associated with increased SCC, a widely used indicator of udder inflammation and milk quality (Wever and Emanuelson, 1989). Pathogens such as *S. dysgalactiae* and *S. uberis* usually elicit an immune response, resulting in elevated SCC levels, which reflect the severity of the infection (Olde Riekerink et al., 2008). Elevated SCC, in turn, is associated with decreased milk yield, as inflammation reduces the functional capacity of the mammary gland (Schukken et al., 2011). In addition to reducing milk yield, IMI caused by *S. dysgalactiae* and *S. uberis* have been previously associated with a negative impact on milk composition, diminishing overall milk quality (Whist et al., 2007; Kester et al., 2015). These negative effects on milk yield and potentially on milk components underscore the importance of managing infections to mitigate production losses and maintain milk quality in dairy herds.

Although some research has already been conducted on *S. uberis* and *S. dysgalactiae*, our study aims to further investigate their impact, along with that of IMI by other

prevalent microorganisms lacking sufficient representation in the literature on quarter-level SCC, milk yield, and milk components. By employing a longitudinal study design, we aimed to capture the dynamic nature of these infections over time, providing a more comprehensive understanding of their influence on milk production and quality. Moreover, by measuring quarter-level IMI status and outcomes (i.e., quarter milk yield, SCC, and composition), we aimed at describing the effects of these IMI with greater precision, as compared with cow-level estimates.

Another study provided an in-depth investigation of these relationships for *Staphylococcus* spp. (Kurban et al., 2026). Therefore, the primary objective of our paper was to estimate the association between IMI caused by microorganisms other than *Staphylococcus* spp., isolated in pure culture, with the SCC, milk yield (kg/d), lactose, protein, and fat contents at the quarter level. As a secondary objective, we estimated the quarter-level apparent prevalence, incidence, and elimination rates of IMI caused by these microorganisms.

MATERIALS AND METHODS

Study Design

The current study was an observational longitudinal study conducted on commercial dairy farms from December 2018 to March 2020. The recommendations for reporting observational studies were followed throughout the text (STROBE-vet guidelines; Sargeant et al., 2016). The study methodology was described in detail in the first article of this series (Kurban et al., 2026). Briefly, we selected a convenience sample of 5 dairy herds in the Montérégie region (QC, Canada) using automated milking systems (AMS; Lely Astronaut A4 AMS technology). The project was approved by the Animal Care and Use Committee of the Faculty of Veterinary Medicine (protocol 18-Rech-1975).

Cow Selection and Milk Sampling

The cows were selected systematically at the moment of their calving (DIM ≥ 3) and followed with samplings every other week up to 305 DIM. Cows were recruited until a sample of 500 cows was achieved. Only Holstein cows were included. The dairy herds were visited every other week, and at each visit (up to 305 DIM; ~20 sampling times per quarter), 50-mL samples of cisternal milk were collected aseptically and manually from each quarter of the selected cows by 2 graduate students and research technicians. Sampling was conducted during on-farm visits based on a pre-established list of animals. This list was prepared in advance for each visit using

herd records, with cows selected according to the study criteria. No quarters or cows were excluded from the study; however, blind quarters were not sampled. In addition, samples from quarters affected by clinical mastitis on a given sampling were excluded from data analysis. The data extracted from the AMS' software provided the quarter-level milk yield for each milking and the time of the day for the different milking. A detailed description is provided in Kurban et al. (2026).

Analysis of SCC and Milk Components

Determination of SCC was performed by Lactanet (Ste-Anne-de-Bellevue, QC, Canada) using a CombiFoss 7 DC instrument (FOSS Electric Analytical Solution, Hillerød, Denmark). All milk samples were warmed to reach a temperature of 39°C to 40°C before analysis. Fat, protein, and lactose content were analyzed by mid-infrared spectroscopy using a MilkoScan FT6000 (FOSS Electric Analytical Solution, Hillerød, Denmark).

Bacteriological Analysis and Definition of Intramammary Infection

Milk bacteriological culture was performed according to the National Mastitis Council guidelines (Hogan et al., 1999). A detailed description was provided at Kurban et al. (2026). Briefly, a quarter was considered positive for a given microorganism when a quarter milk sample contained ≥ 1 cfu/10 μ L of milk of this species in mixed or pure culture. Based on culture results, milk samples were classified as (1) pure culture (one type of microorganism colony), (2) mixed culture (2 different types of microorganism colonies), (3) contaminated sample (≥ 3 different types of microorganism colonies), or (4) no growth (if no growth was observed after 48 h). For determination of the IMI status of the quarters, bacteriological results were used as follows:

- a quarter was considered healthy only in case of no growth results;
- otherwise, a quarter was considered to be infected with microorganisms other than *Staphylococcus* spp. that were previously reported in the literature as associated with subclinical or clinical mastitis (Kurban et al., 2022). The microorganisms that were isolated at least 30 times throughout the study period were *Aerococcus viridans*, *Bacillus pumilus*, *Bacillus licheniformis*, *Lactococcus garvieae*, *Streptococcus uberis*, *Streptococcus dysgalactiae*, *Corynebacterium bovis*, *Corynebacterium amycolatum*, *Corynebacterium xerosis*, *Escherichia coli*, *Mammaliicoccus sciuri*, and *Klebsiella pneumoniae*.

- The classification “other IMI” was assigned if the bacteriological culture identified *Staphylococcus* spp. or any other microorganism isolated fewer than 30 times throughout the study period.

A complete list of all microorganisms identified by the microbiological culture was provided in Supplemental Material S1 and S2 (see Notes).

Data

Data were checked for missing and unusual values before analysis. The inclusion criteria were (1) records from quarters within 3 to 305 DIM, and (2) cows should be Holstein breed. For the milk content, the calibration range values obtained from FOSS were as follows: 1.44% to 10.38% for fat, 2.59% to 4.60% for protein, and 3.76% to 5.53% for lactose. Parameters from a given milk sample with values outside of the MilkoScan calibration range were recorded as missing information. Given that cows on AMS are free to be milked whenever they choose, the number of milkings on a given calendar day is variable. Therefore, to compute the daily milk yield per quarter, we added the preceding 3-d production of a quarter (in kg) and divided the total by 3. Thus, the daily milk yield (in kg/d) was a 3-d average milk yield. The dataset was divided into primiparous and multiparous cows for analysis.

Case Definition

As an arbitrary decision, after applying the case definitions below, we retained for analysis only microorganisms that were isolated at least 30 times throughout the lactation period. In analysis 2, all microorganisms isolated fewer than 30 times were grouped under “other IMI.”

Liberal Case Definition. To estimate the apparent prevalence, the incidence rate, and the elimination rate of an IMI, we applied a liberal case definition. In this liberal case definition, we used one sample to determine the quarter's IMI status at a given day. Similarly, for IMI elimination, only one negative sample after a positive sample for a given microorganism was required to consider the infection resolved.

Strict Case Definition. To estimate the association of IMI with SCS, milk yield, lactose, protein, and fat, we used a stricter case definition. This definition used 2 consecutive samples to determine the quarter's IMI status. For instance, a quarter was considered to be infected with a given species only if 2 consecutive samples tested positive for that same given species in pure culture. Similarly, for a quarter to be considered healthy, 2 consecutive no-growth samples were required. Only the observations that met the case definition criteria were retained for

further analysis. For example, the IMI status was deemed missing if any of the samples were contaminated or missing, or if there were contradictory results between the 2 samples (e.g., a positive result followed by a negative one, or vice versa). This choice was made to increase the probability, in the analyses evaluating the impact of IMI on milk yield, SCS, and milk components, that quarters classified as infected or healthy were truly infected or healthy. Nevertheless, as part of a sensitivity analysis, we also reran these models using the liberal case definition (provided as Supplemental Material S8, see Notes).

Statistical Analysis

Apparent Prevalence, Incidence Rate, and Elimination Rate of IMI (Analysis 1). In our initial analyses, using generalized mixed models with random intercepts for quarter, cow, and herd resulted in nonconvergence. We then attempted to use generalized estimating equations for modeling but did not achieve success for all models with this approach either. The only modeling method that consistently worked well was applying robust variance to account for clustering, and all results reported were, therefore, derived from these models. To estimate the prevalence, incidence rate, and elimination rate for a given IMI, we first ran a model without any predictor. The inverse logit function was then applied to the obtained intercept to obtain the prevalence, incidence, or elimination rate. Moreover, we also estimated the prevalence, incidence rate, and elimination rate as function of DIM. For simplicity, for these latter analyses, we kept DIM in the model even if not significant. For clarity, as we used a frequentist approach, the estimates reported should be interpreted as apparent prevalence, apparent incidence rate and apparent elimination rate (i.e., assuming microbiological culture is a perfect test).

Associations of IMI with SCS, Milk Yield, and Milk Components (Analysis 2). Given the hierarchical structure of the data, a linear mixed model was developed with herd, cow, and quarter as random intercepts, using a restricted maximum likelihood estimator with SCS, milk yield, lactose, protein, or fat as the outcome variables. Due to the repeated measures within a quarter throughout the lactation period (up to 20 records per quarter), an exponential correlation structure was incorporated into the models (spatial covariance structure). The linearity of the relationship between the outcomes and DIM was visually assessed using scatter plots with smoothed curves. Polynomial models were explored to further investigate the linearity assumption. When the linearity assumption was not satisfied, the DIM variable was retained in polynomial form. We initially ran models that included only the IMI variable (coded as indicated in Table 1) to estimate its association with each outcome. Subsequently, we ran

another model that incorporated the interaction between IMI and DIM. If the interaction term was not significant, we reran the models but, at that time, including DIM as a covariate. For clarity and consistency, DIM was retained in these models as a covariate regardless of its significance. Data for SCC were transformed to SCS using the following formula (Shook, 1993):

$$SCS = \log_2 \left(\frac{SCC}{100} \right) + 3,$$

where SCC is given in cells per microliter.

The post hoc Tukey method was used to adjust for multiple comparisons for the IMI status variable.

Model Evaluation. Best linear unbiased predictions were calculated for random effects and were plotted against the predicted outcome to verify the assumption of homoscedasticity. Normality assumptions were verified by plotting the quantiles of the BLUP for both random effects and the lowest level residuals against quantiles of normal distribution. Statistical analyses were performed using RStudio version 2024.09.0 (<https://posit.co/>).

Reproducibility. The dataset, executable R script, and supplemental material are available at <https://doi.org/10.5683/SP3/TDOBTR>.

RESULTS

Data

Initially, a total of 21,580 milk sample records from the lactational period were obtained for the larger project on staphylococci IMI. During the data cleaning and validation process, we identified 2 quarters with an unusual sampling series. The corresponding milk records for these quarters were dropped, as they belonged to blind quarters that were inadvertently sampled on a few occasions. An additional 111 observations were excluded because they fell outside the specified DIM range. Ultimately, a total of 21,467 milk sample records were available for data analysis (Supplemental Material S1 and S2).

Primiparous Cows. Initially, 4,601 quarter milk sample records were available for analysis. The final number of observations used for each model in analysis 1 is fully reported in Supplemental Material S5 (see Notes). For analysis 2, a total of 389 milk records were excluded due to mixed culture or contaminated results. The final available dataset for first lactation cows included 4,212 milk records from 319 quarters of 80 cows within the 5 herds. The final number of milk records included in the dataset for the second analysis were as follows: 3,204 for SCS, 4,133 for milk yield, 2,768 for lactose, 2,779 for protein, and 2,610 for fat. These differences were due to missing values for some of these parameters.

Table 1. Number of records and proportion of pathogens identified using bacterial culture from milk samples collected from 5 dairy herds in Montérégie region (QC, Canada) as well as descriptive statistics for SCS, milk yield (kg/d), lactose (%), protein (%), and fat (%) during the lactation period

Independent variable	Records (n)	% of total	SCS		Milk yield (kg/d)		Lactose (%)		Protein (%)		Fat (%)	
			Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
IMI Status												
Primiparous cows ¹												
No growth	3,023	71.8	1.6	1.6	7.8	2.4	4.6	0.3	3.2	0.3	4.6	2.2
<i>Aerococcus viridans</i>	111	2.6	1.2	1.2	8.4	2.4	4.7	0.3	3.2	0.3	4.0	1.9
<i>Corynebacterium bovis</i>	37	0.9	2.2	1.5	9.4	2.2	4.5	0.3	3.2	0.2	4.0	2.1
<i>Streptococcus dysgalactiae</i>	31	0.7	7.5	2.5	4.2	2.1	4.4	0.2	3.3	0.4	3.0	1.2
Other IMI ²	1,010	24.0	3.0	2.3	7.9	2.4	4.6	0.3	3.2	0.3	4.3	2.0
Multiparous cows ³												
No growth	6,014	76.5	1.8	1.8	10.3	3.4	4.4	0.3	3.2	0.3	5.1	2.4
<i>Streptococcus dysgalactiae</i>	225	2.9	7.5	1.6	9.6	3.8	4.3	0.3	3.3	0.2	4.5	1.9
<i>Corynebacterium bovis</i>	191	2.4	2.6	1.7	10.2	3.6	4.4	0.3	3.1	0.3	4.7	2.3
<i>Streptococcus uberis</i>	107	1.4	7.4	1.9	10.7	5.0	4.2	0.3	3.2	0.3	4.0	1.5
<i>Corynebacterium amycolatum</i>	69	0.9	2.7	1.6	11.2	3.5	4.4	0.3	3.2	0.3	4.4	2.4
<i>Lactococcus garvieae</i>	37	0.5	3.9	1.6	8.6	3.9	4.4	0.3	3.2	0.4	4.2	1.9
Other IMI ²	1,218	15.4	5.4	2.1	10.0	3.6	4.3	0.3	3.2	0.3	4.8	2.2

¹For primiparous cows, the reported descriptive results were derived from the liberal case definition (sensitivity analysis).

²Other IMI includes IMI caused by *Staphylococcus* sp. or IMI caused by microorganisms that were isolated fewer than 30 times throughout the lactation period.

³For multiparous cows, the reported descriptive results were derived from the conservative case definition. The results for the liberal case definition (sensitivity analysis) are fully reported in the Supplemental Material S8 (see Notes).

Multiparous Cows. Initially, 16,866 quarter milk sample records were available for analysis. The final number of observations used for each model in analysis 1 is fully reported in Supplemental Material S6 (see Notes). For analysis 2, a total of 9,005 records were excluded due to mixed cultures, contamination, or not meeting the conservative case definition criteria (Supplemental Material S2). The final available dataset for cows with 2 or more lactations included 7,861 quarter milk sample records from 1,131 quarters of 299 cows within the 5 herds. The final number of milk records included in the dataset for the second analysis were as follows: 5,939 for SCS, 7,664 for milk yield, 4,870 for lactose, 5,247 for protein, and 4,703 for fat. These differences were due to missing values for some of these parameters.

Descriptive Statistics

The number of lactating cows per herd ranged from 60 to 380, with a mean (SD) milk production (kg/cow per day) ranging from 34.3 (10.0) to 43.6 (10.3). The mean (SD) SCS ranged from 2.2 (2.2) to 2.9 (2.6), and the mean (SD) quarter milk yield (kg/d) for each herd ranged from 8.9 (3.0) to 11.3 (3.2). The complete demographic information on the selected herds was reported by Kurban et al. (2026).

Primiparous Cows. At the quarter level, 71.8% (3,023/4,212) of the quarters were deemed healthy on testing day, 24.0% (1,010/4,212) were infected with other IMI, 2.6% (111/4,212) were infected with *A. viridans*,

0.9% (37/4,212) were infected with *C. bovis*, and 0.7% (31/4,212) were infected with *S. dysgalactiae*. The mean (SD) for SCS, milk yield (kg/d), lactose (%), protein (%), and fat (%) was 2.0 (1.9), 7.8 (2.4), 4.6 (0.3), 3.2 (0.3), and 4.5 (2.1), respectively. The mean (SD) DIM was 122.3 (77.6), with an interquartile range (IQR) of 133. The complete descriptive statistics are reported in Table 1.

Multiparous Cows. At the quarter level, 76.5% (6,014/7,861) of the quarters were deemed healthy, 15.4% (1,218/7,861) were infected with other IMI, 2.9% (225/7,861) were infected with *S. dysgalactiae*, 2.4% (191/7,861) were infected with *C. bovis*, 1.4% (107/7,861) were infected with *S. uberis*, 0.9% (69/7,861) were infected with *C. amycolatum*, and 0.5% (37/7,861) were infected with *L. garvieae*. The mean (SD) for SCS, milk yield (kg/d), lactose (%), protein (%), and fat (%) was 2.6 (2.4), 10.2 (3.4), 4.4 (0.3), 3.2 (0.3), and 5.0 (2.4), respectively. The mean (standard deviation) DIM was 113.0 (71.3), with an IQR of 115. The complete descriptive statistics are reported in Table 1.

Apparent Prevalence, Incidence Rate, and Elimination Rate of IMI

Primiparous Cows. Three microorganisms met the criteria for the liberal case definition: *A. viridans*, *C. bovis*, and *S. dysgalactiae* (Supplemental Material S5). The mean (95% CI) prevalence of IMI caused by *A. viridans*, *S. dysgalactiae*, and *C. bovis* were 3.8 (3.3, 4.4), 0.9 (0.7,

1.3), and 0.9 (0.7, 1.2), respectively. The prevalence of *A. viridans* was generally higher during the first 60 DIM, whereas *C. bovis* and *S. dysgalactiae* showed higher prevalence between 150 and 250 DIM (Figure 1). Among these 3 microorganisms, *A. viridans* had the highest mean incidence rate; however, infections caused by *A. viridans* also exhibited a higher mean elimination rate compared with those caused by *S. dysgalactiae* (Figure 1 and Table 2). For instance, an incidence of 1.8 new cases of IMI caused by *A. viridans* was observed per 100 quarters per week, with 46.0% of these infections likely to be eliminated within one week. In contrast, *S. dysgalactiae* exhibited a lower incidence of only 0.1 new cases per 100 quarters per week; however, only ~4.0% of these infections were expected to be eliminated within one week. The prevalence, incidence rate, and elimination rate as function of DIM can be found in Figure 1.

Multiparous Cows. Twelve microorganisms met the criteria for the liberal case definition: *A. viridans*, *B. lincheniformis*, *B. pumilus*, *C. amycolatum*, *C. bovis*,

C. xerosis, *E. coli*, *K. pneumoniae*, *L. garvieae*, *M. sciuri*, *S. dysgalactiae*, and *S. uberis* (Supplemental Material S6). The 3 most prevalent microorganisms were *C. bovis*, *A. viridans*, and *S. dysgalactiae*, with mean prevalences (95% CI) of 5.2% (4.8%, 5.5%), 3.7% (3.4%, 4.0%), and 2.8% (2.5%, 3.1%), respectively (Table 2). The prevalence of *A. viridans* remained consistent throughout lactation, whereas *C. bovis* exhibited higher prevalence between 150 and 250 DIM, and *S. dysgalactiae* was more prevalent in the early stages of lactation (Figure 2). Although *A. viridans* had the highest mean incidence rate, it also had one of the highest mean elimination rates (Table 2). The lowest elimination rate was observed for *S. dysgalactiae*, followed by *S. uberis* and *L. garvieae* (Table 2). For instance, similar to what was observed for primiparous cows, an expected incidence of 1.7 new cases of IMI caused by *A. viridans* was observed per 100 quarters per week, with 46.0% of these infections likely to be eliminated within one week. In contrast, *S. dysgalactiae*, *S. uberis*, and

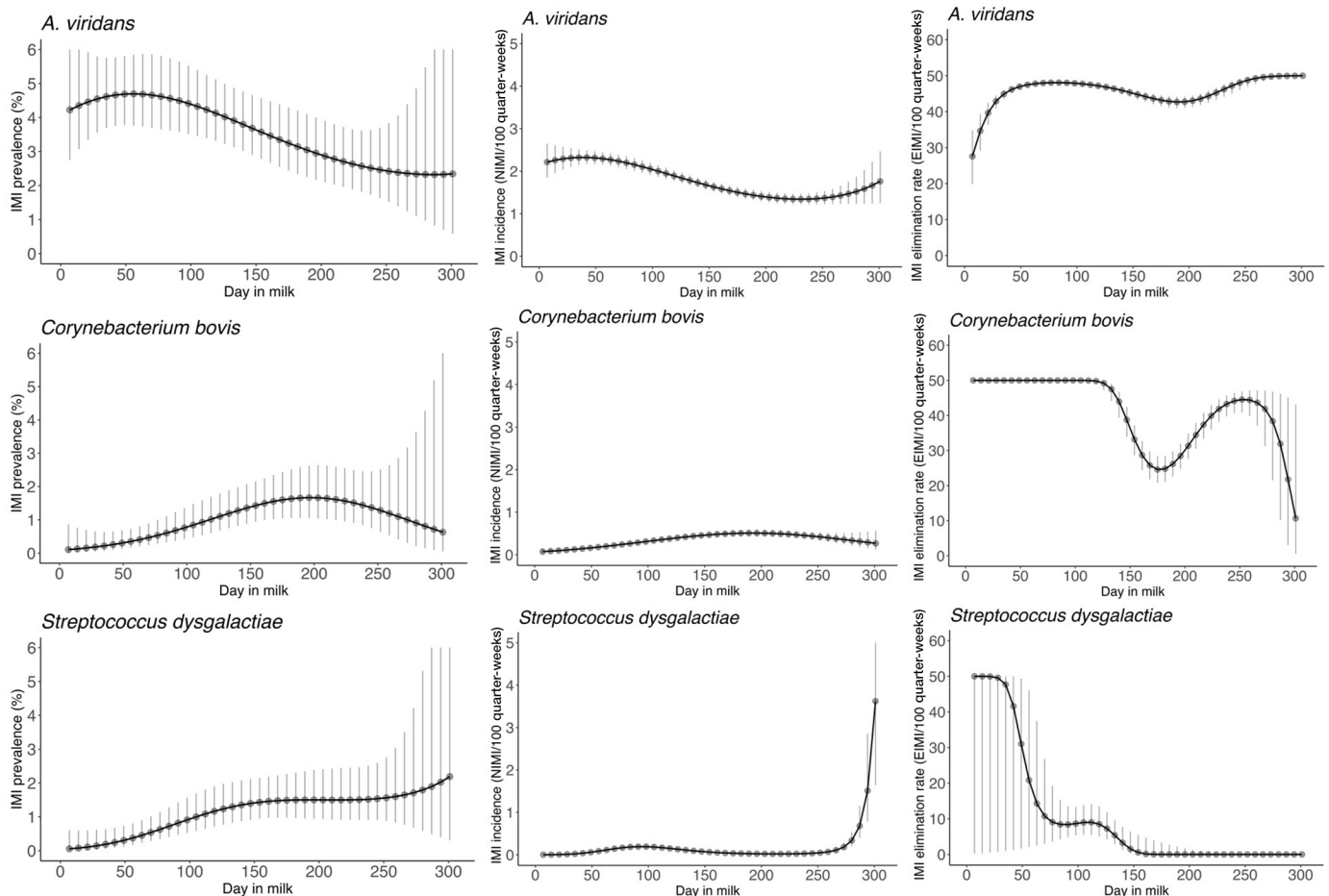


Figure 1. Prevalence (%), incidence rate (new intramammary infections/100 quarter-weeks), and elimination rate (elimination of intramammary infections/100 quarter-weeks) of intramammary infections as a function of DIM, for pathogens isolated from primiparous cows, from dairy herds in Québec. Error bars represent the 95% CI.

Table 2. Prevalence (%), incidence rate (new intramammary infections/100 quarter-weeks), and elimination rate (elimination of intramammary infections/100 quarter-weeks) of intramammary infections caused by pathogens previously associated with subclinical or clinical mastitis from Québec dairy herds

Pathogen	Prevalence		Incidence rate		Elimination rate	
	Mean	95% CI	Mean	95% CI	Mean	95% CI
Primiparous						
<i>Aerococcus viridans</i>	3.8	3.3, 4.4	1.8	1.6, 2.2	45.8	43.1, 47.6
<i>Corynebacterium bovis</i>	0.9	0.7, 1.2	0.3	0.2, 0.5	37.2	28.8, 43.1
<i>Streptococcus dysgalactiae</i>	0.9	0.7, 1.3	0.1	0.04, 0.2	3.9	1.3, 10.8
Multiparous						
<i>Corynebacterium bovis</i>	5.2	4.8, 5.5	1.7	1.6, 1.9	32.4	30.5, 34.2
<i>Aerococcus viridans</i>	3.7	3.4, 4.0	1.7	1.5, 1.8	46.0	44.6, 47.0
<i>Streptococcus dysgalactiae</i>	2.8	2.5, 3.1	0.2	0.2, 0.3	8.3	6.6, 10.4
<i>Streptococcus uberis</i>	1.2	1.0, 1.4	0.1	0.1, 0.2	11.4	8.5, 14.9
<i>Corynebacterium amycolatum</i>	1.2	1.0, 1.4	0.4	0.3, 0.5	27.8	23.9, 31.6
<i>Escherichia coli</i>	0.7	0.6, 0.8	0.3	0.2, 0.4	40.9	36.3, 44.1
<i>Lactococcus garvieae</i>	0.6	0.5, 0.7	0.1	0.1, 0.2	13.0	8.4, 19.0
<i>Bacillus licheniformis</i>	0.4	0.3, 0.5	0.2	0.2, 0.3	47.4	40.2, 49.4
<i>Corynebacterium xerosis</i>	0.4	0.3, 0.5	0.2	0.2, 0.3	N/A ¹	N/A
<i>Mammaliicoccus sciuri</i>	0.4	0.3, 0.5	0.1	0.1, 0.2	36.7	29.6, 41.9
<i>Klebsiella pneumoniae</i>	0.3	0.2, 0.4	0.1	0.1, 0.2	33.8	25.9, 40.1
<i>Bacillus pumilus</i>	0.3	0.2, 0.4	0.2	0.1, 0.2	50.0	50.0, 50.0

¹The model failed to converge due to the complete elimination of all intramammary infections. N/A = not applicable.

L. garvieae showed lower mean incidences, with only 0.2, 0.1, and 0.1 new cases per 100 quarters per week, respectively. However, only 8.3% of *S. dysgalactiae* infections, 11.4% of *S. uberis* infections, and 13.0% of *L. garvieae* infections were expected to be eliminated within one week. The prevalence, incidence rate, and elimination rate for all microorganisms as a function of DIM can be found in Figure 2.

Associations of IMI with SCS, Milk Yield, and Milk Components

Primiparous Cows. For first-lactation cows, it was not possible to apply the conservative case definition because no bacterial species were isolated at least 30 times during the sampling period under this definition (Supplemental Material S1). Thus, for first-lactation cows, we had to use the liberal case definition (Supplemental Material S7, see Notes) to report the results. Three microorganisms met the liberal case definition: *A. viridans*, *C. bovis*, and *S. dysgalactiae*. The interaction between IMI and DIM was significant only when SCS and lactose were considered as the outcome. Additionally, protein and fat content were not associated with the IMI status of a quarter (Table 3 and Supplemental Material S7). Although the IMI status of a quarter was not statistically associated with fat content (%), quarters infected with *S. dysgalactiae* had a numerical decrease of ~1% compared with healthy quarters (Table 3).

The association of the IMI status of a quarter with the SCS depended on the DIM, and a greater difference in SCS between quarters infected with *S. dysgalactiae* and

healthy quarters was observed between ~120 and 210 DIM (Figure 3). For instance, at 150 DIM, the mean (95% CI) SCS for a healthy quarter was 1.7 (1.3, 2.1), whereas for a quarter infected with *S. dysgalactiae*, it was 7.4 (5.7, 9.2). In addition, quarters infected with *S. dysgalactiae* also had a higher mean SCS compared with quarters infected with *A. viridans* or *C. bovis* (Table 3 and Supplemental Material S7). The mean SCS from quarters infected with *A. viridans* or *C. bovis* did not differ from healthy quarters.

When considering milk yield as the outcome, the association with IMI status was constant across DIM. Quarters infected with *S. dysgalactiae* produced 1.3 kg/d less milk compared with healthy quarters (Table 3 and Figure 3). In addition, quarters infected with *S. dysgalactiae* also produced less milk than quarters infected with *A. viridans* or *C. bovis* (Table 3). The milk yield from quarters infected by *A. viridans* or *C. bovis* did not differ from healthy quarters.

Regarding the lactose content, the association with the IMI status of a quarter depended on the DIM, although the magnitude of these changes was very small. For instance, at 150 DIM, the lactose content from quarters infected with *S. dysgalactiae*, *A. viridans*, or *C. bovis* did not differ from healthy quarters (Table 3). In comparison, at 210 DIM, quarters infected with *S. dysgalactiae* had a decrease of 0.3% compared with healthy quarters, whereas quarters infected with *C. bovis* had an increase of 0.2% (Figure 3). The association of IMI with SCS, milk yield, and lactose for all microorganisms as function of DIM can be found in Figure 3. All models are fully reported in Supplemental Material S7.

Multiparous Cows. For cows with 2 or more lactations, 5 microorganisms met the conservative case definition: *C. amycolatum*, *C. bovis*, *L. garvieae*, *S. dysgalactiae*, and *S. uberis* (Supplemental Material S2 and S9, see Notes). The interaction between IMI and DIM was not significant when fat was considered as the outcome. Moreover, no significant association was observed between milk yield or protein content with the IMI status of a quarter (Table 4 and Supplemental Material S9). Although milk yield was not statistically associated with IMI status, quarters infected with *S. dysgalactiae* produced 0.7 kg/d less (numerical difference) compared with healthy quarters (Table 4).

The association between a quarter's IMI status and its SCS varied depending on the DIM. For instance, at 150 DIM, quarters infected with *S. dysgalactiae* and *S. uberis* had a mean (95% CI) SCS of 7.3 (6.5, 8.2) and 6.5 (5.5, 7.5), respectively, whereas healthy quarters had a mean (95% CI) SCS of 1.9 (1.6, 2.1; Table 4). In addition, quarters infected with *S. dysgalactiae* and *S. uberis*

had a higher mean SCS compared with quarters infected with *C. amycolatum*, *C. bovis*, or *L. garvieae* (Table 4 and Supplemental Material S9). No statistical difference was observed in the mean SCS for quarters infected with *S. dysgalactiae* and *S. uberis*.

The association between a quarter's IMI status and lactose content varied with DIM. As observed in primiparous cows, these changes were minimal. For example, the lactose content (%) was slightly reduced in quarters infected with *S. dysgalactiae*, with a decrease of 0.3% at 150 DIM compared with healthy quarters. The association between a quarter's IMI status and fat content remained constant across DIM. For instance, compared with healthy quarters, quarters infected with *S. uberis* showed the greatest reduction in fat content (−1.0%), followed by *C. bovis* (−0.8%; Table 4). The association of IMI status with SCS and lactose for all microorganisms as function of DIM can be found in Figure 4.

As a sensitivity analysis, we reran the models using the liberal case definition. The results remained consistent

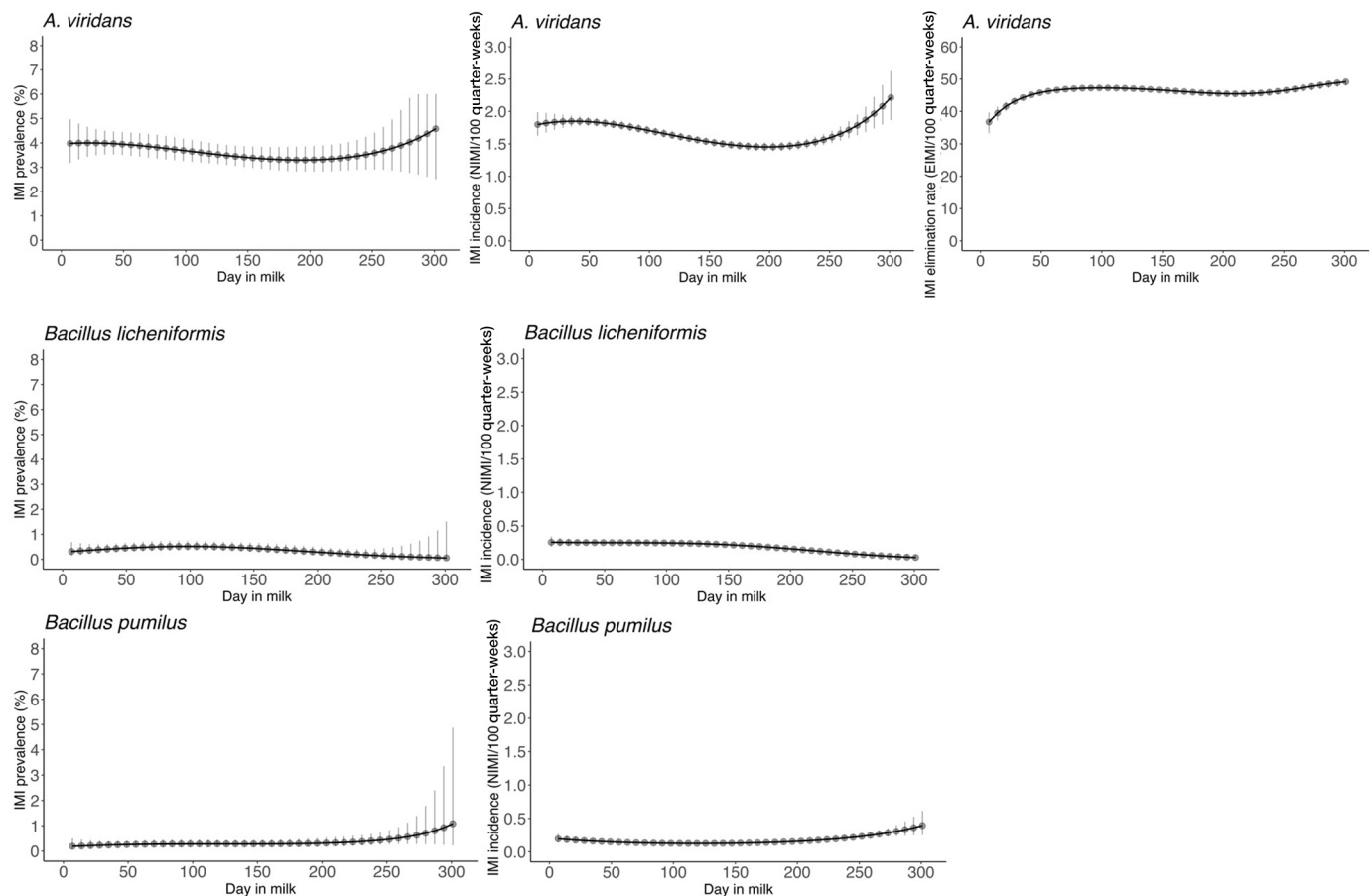


Figure 2. Prevalence (%), incidence rate (new intramammary infections/100 quarter-weeks), and elimination rate (elimination of intramammary infections/100 quarter-weeks) of intramammary infections as a function of DIM, for pathogens isolated from multiparous, from dairy herds in Québec. If a pathogen does not have a figure for the elimination rate, it indicates that all infections were resolved. Error bars represent the 95% CI.

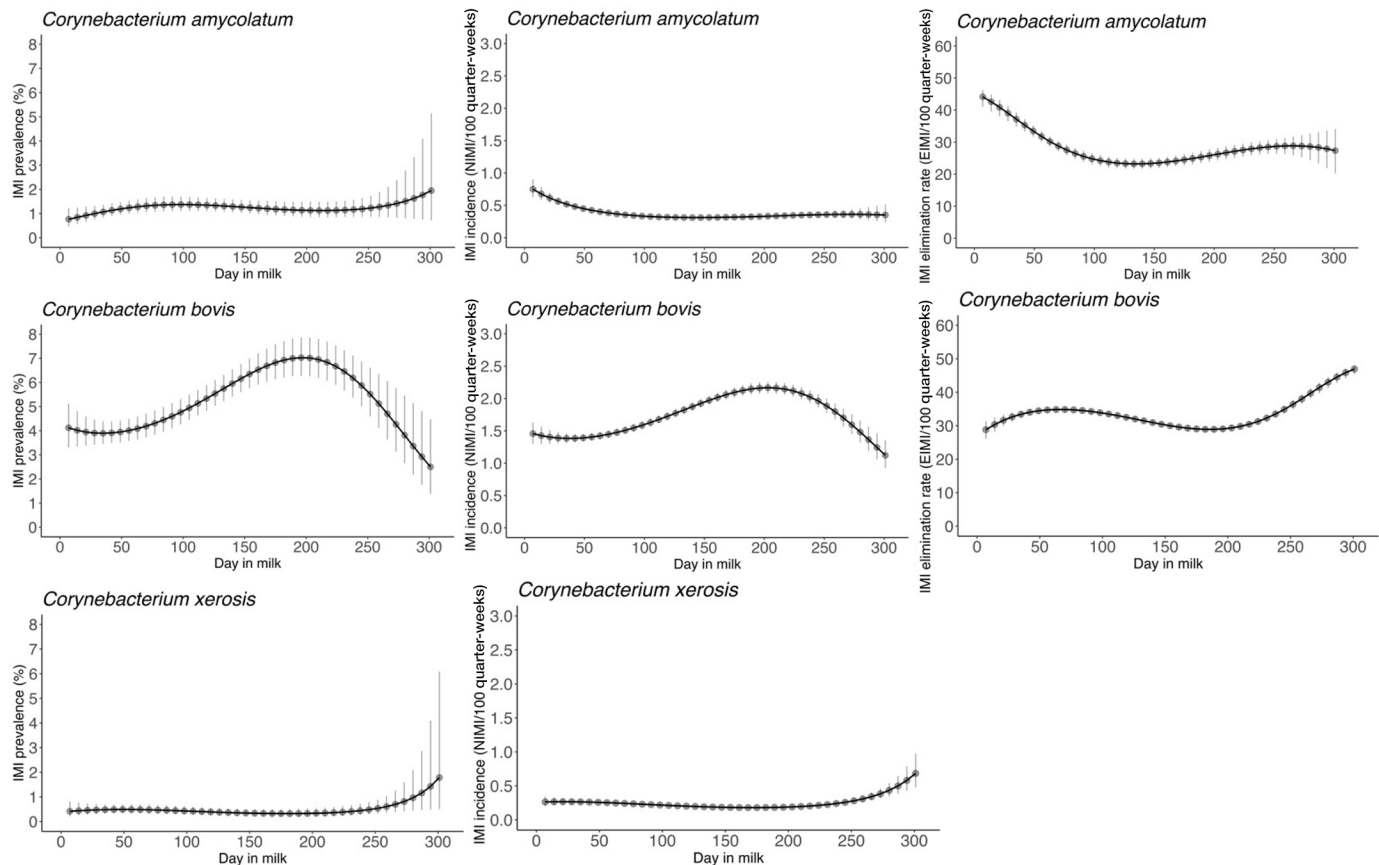


Figure 2 (Continued). Prevalence (%), incidence rate (new intramammary infections/100 quarter-weeks), and elimination rate (elimination of intramammary infections/100 quarter-weeks) of intramammary infections as a function of DIM, for pathogens isolated from multiparous, from dairy herds in Québec. If a pathogen does not have a figure for the elimination rate, it indicates that all infections were resolved. Error bars represent the 95% CI.

with the conservative approach, identifying *S. uberis* and *S. dysgalactiae* as the microorganisms associated with the most significant increases in SCS without any correlated milk loss. Additionally, under this case definition, more microorganisms met the criteria, with *E. coli* showing a mean SCS of 3.6 and a milk loss of 1.2 kg/d compared with healthy quarters. The complete results for the liberal case definition (sensitivity analysis) are reported in Supplemental Material S8.

DISCUSSION

The primary objectives of this study were to estimate the incidence and elimination rate of IMI caused by microorganisms other than *Staphylococcus* spp. in dairy herds in Québec, as well as to describe how these epidemiological indices varied throughout the lactation period. Additionally, the study aimed to investigate the associations of these IMI with SCS, milk yield, and milk components.

In our study, the mean SCS was 2.0 (i.e., SCC = 50,000 cells/mL) for primiparous cows and 2.6 (i.e., SCC = 75,786 cells/mL) for multiparous cows, which is consistent with values reported in other studies on Holstein cows (Bobbo et al., 2016; Franzoi et al., 2020; Pegolo et al., 2021). The mean quarter milk yield increased with parity, as expected. Primiparous cows produced an average of 7.8 kg/quarter-day, whereas multiparous cows averaged 10.2 kg/quarter-day, which aligns with previous studies on dairies using AMS. For instance, Penry et al. (2018) reported a mean quarter milk yield of ~3.2 kg/milking for Holstein cows, and Silva Boloña et al. (2022) reported a similar mean of 3.1 kg/milking. Given the average milking interval of 8 to 9 h in these studies, the daily quarter milk yield in kilograms is comparable to our findings. Additionally, the mean values for protein and lactose contents were similar between primiparous and multiparous cows and consistent with previous studies on Holstein cows (Schwarz et al., 2020a; Pegolo et al., 2021; Bisutti et al., 2022; Huang et al., 2023). The fat content

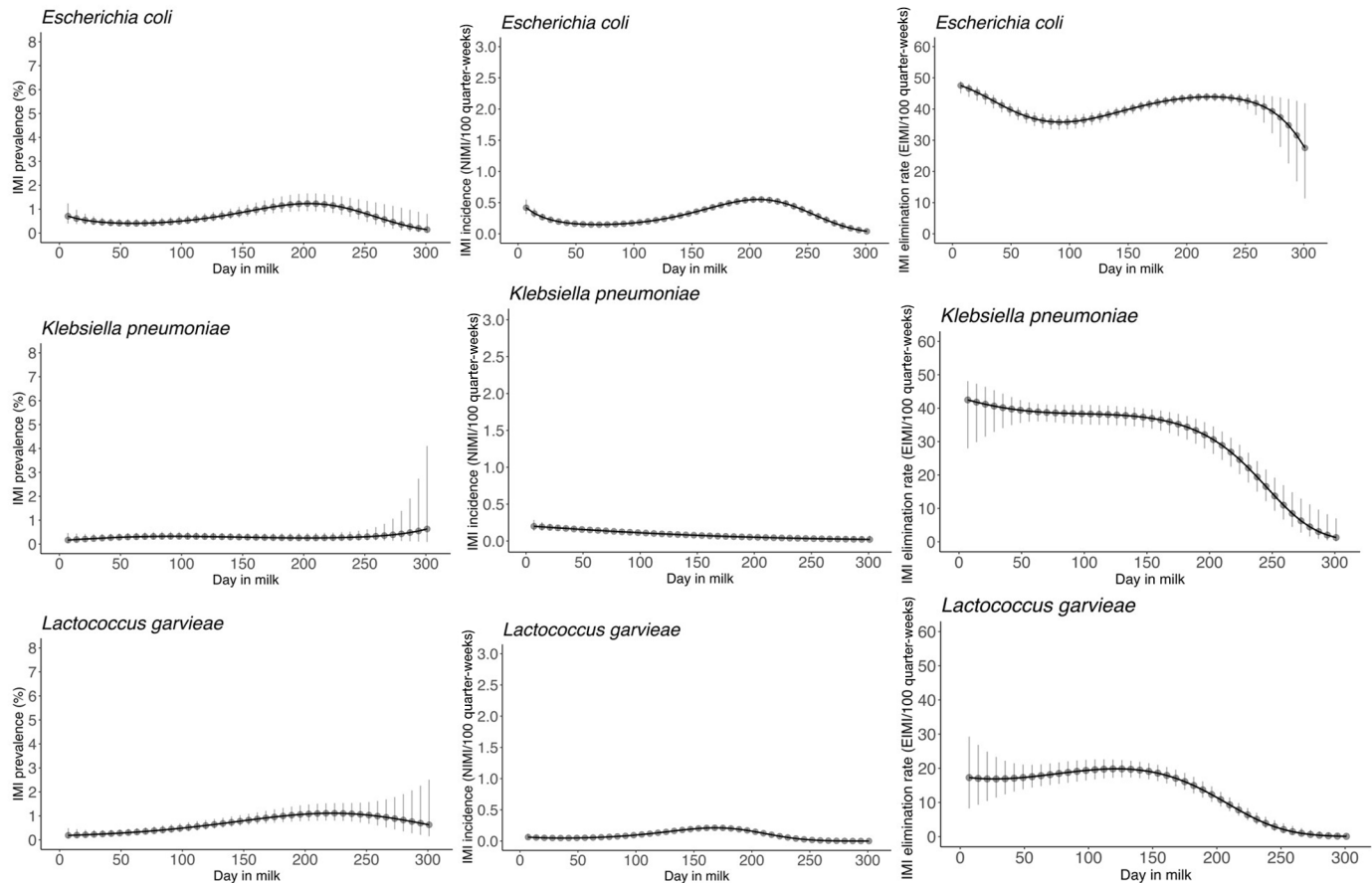


Figure 2 (Continued). Prevalence (%), incidence rate (new intramammary infections/100 quarter-weeks), and elimination rate (elimination of intramammary infections/100 quarter-weeks) of intramammary infections as a function of DIM, for pathogens isolated from multiparous, from dairy herds in Québec. If a pathogen does not have a figure for the elimination rate, it indicates that all infections were resolved. Error bars represent the 95% CI.

(%) in multiparous cows in our study was slightly higher than reported in other studies on Holstein cows (Schwarz et al., 2020a; Pegolo et al., 2021; Huang et al., 2023). In a previous analysis using the same dataset, we found that quarters with high SCS were associated with a higher fat percentage (Fonseca et al., 2025). However, Huang et al. (2023) reported lower fat yields (in kg/d) in cows with high SCS. These findings suggest that the higher fat proportion observed in our study may be due to a concentration effect, as quarters with high SCS had a reduced milk yield, leading to an increased fat percentage.

Among the microorganisms isolated from milk in this study, *A. viridans*, *C. bovis*, and *S. dysgalactiae* were the most prevalent in both primiparous and multiparous cows. Our results align with a recent meta-analysis that reported these microorganisms among the 20 most isolated from apparently healthy quarters from 50,429 milk samples from Canada, the United States, and Brazil (Kurban et al., 2022). Our study also revealed that *A. viridans* had the highest incidence rate, although 46%

of these IMI were cleared within one week. In contrast, our results suggested that IMI caused by *S. dysgalactiae* required a longer time for elimination, with only 3.9% of the infections resolving within one week in primiparous cows and 8.3% in multiparous cows. Similar findings were observed for IMI caused by *S. uberis* in multiparous cows, where only 11.4% of infections were expected to resolve within one week. Our results aligns with the results from the study by Woudstra et al. (2023), which reported that the probability of a quarter remaining infected by *S. dysgalactiae* for more than 84 d was 58.7%. This same study also reported that the probability of a quarter infected with *S. uberis* remaining infected for more than 70 d was 53.5% (Woudstra et al., 2023). In this same longitudinal study, the authors also explored the strain diversity of IMI caused by *S. dysgalactiae* and *S. uberis*. They found that ~50% of *S. dysgalactiae* IMI were caused by a single strain, with another 33% attributed to just 2 other strains. These findings suggest that contagious transmission may have played a significant

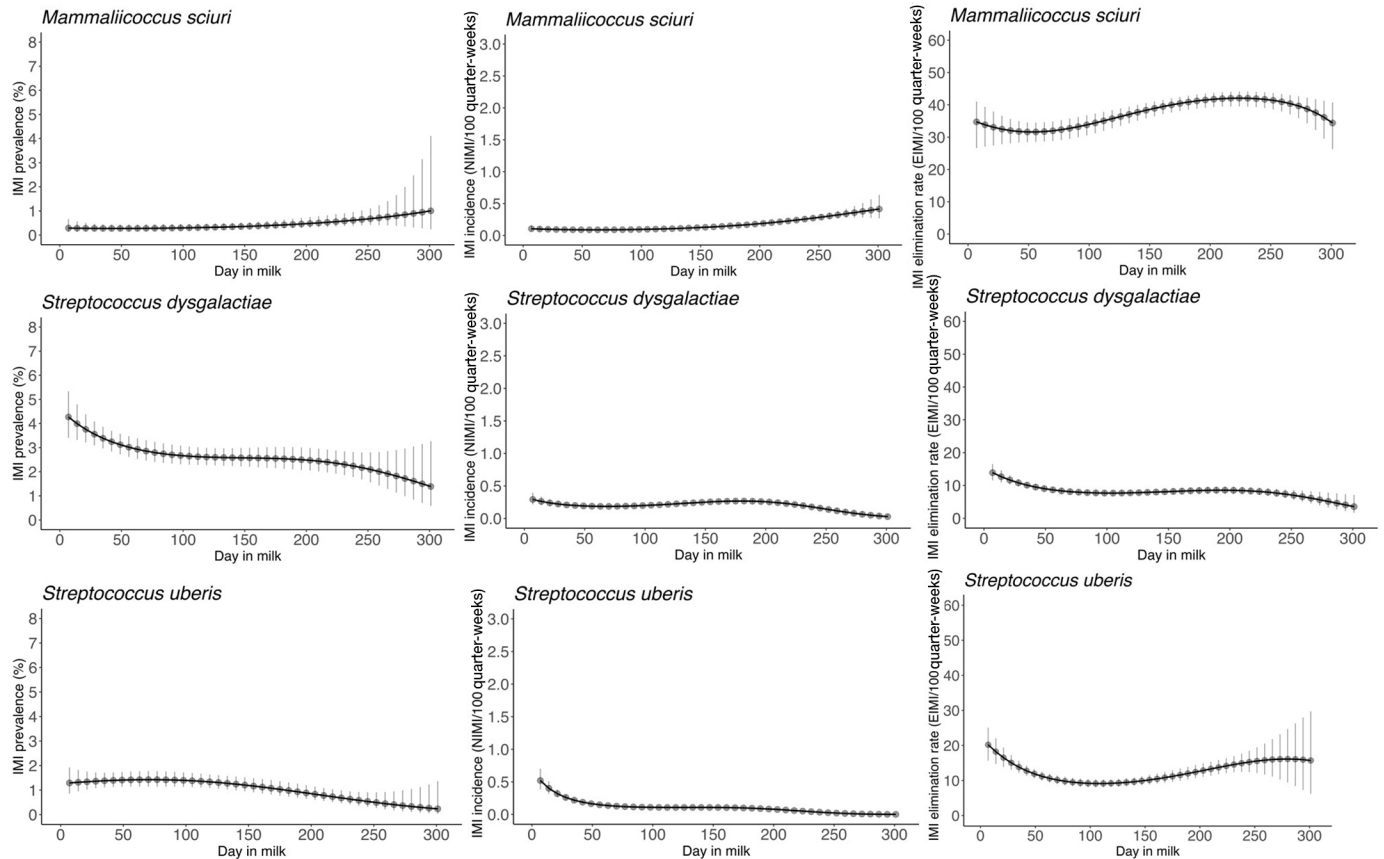


Figure 2 (Continued). Prevalence (%), incidence rate (new intramammary infections/100 quarter-weeks), and elimination rate (elimination of intramammary infections/100 quarter-weeks) of intramammary infections as a function of DIM, for pathogens isolated from multiparous, from dairy herds in Québec. If a pathogen does not have a figure for the elimination rate, it indicates that all infections were resolved. Error bars represent the 95% CI.

role in the dynamics of *S. dysgalactiae* infections. In contrast, although a greater variety of strains was identified for *S. uberis*, the presence of dominant strains still indicated the potential influence of contagious transmission (Woudstra et al., 2023). Additionally, although *S. uberis* is commonly found in the environment, it is increasingly recognized that contagious transmission may play a role in its spread (Davies et al., 2016). It is important to note that the incidence of new infections and the assumptions regarding infection elimination in our study were based exclusively on our case definition, as strain typing was not performed. Consequently, our results do not allow us to distinguish between persistent infections caused by the same strain and new infections involving different strains.

It is common practice to categorize mastitis pathogens into major and minor groups, with IMI caused by major pathogens such as *S. dysgalactiae* and *S. uberis* being associated with high SCC (Sharma et al., 2011; Schwarz et al., 2020b). In our study, we found that IMI in primiparous cows caused by *S. dysgalactiae* were associ-

ated with higher SCS compared with a healthy quarter, and the mean SCS for those infections was higher than for IMI caused by *C. bovis* and *A. viridans*. Similarly, IMI caused by *S. dysgalactiae* and *S. uberis* in multiparous cows were associated with high SCS compared with healthy quarters, and these IMI had a higher mean SCS compared with those caused by minor pathogens. These findings are consistent with previously reported literature. For instance, Zeconi et al. (2021) reported that in the first 30 d of lactation, the mean SCS values were 5.3 and 5.2 for IMI caused by *S. dysgalactiae* and *S. uberis*, respectively, whereas quarters with no bacterial growth had a mean SCS of 4.5. In this same study, they also reported a higher mean SCS for major pathogens compared with minor pathogens (Zeconi et al., 2021).

Intramammary infections caused by minor pathogens are often associated with an elevated SCS, though this increase is typically smaller compared with infections caused by major pathogens. In our study, IMI caused by minor pathogens such as *C. bovis*, *C. amycolatum*, and *L. garvieae* resulted in a numerically higher SCS than

Table 3. Marginal estimates derived from the final mixed effects models for explaining the variation in the quarter SCS, milk yield (kg/d), and lactose (%) during the lactation period according to the IMI status of primiparous cows

Item	Marginal estimate	SE	95% CI
SCS as outcome ¹			
IMI			
No growth (healthy)	1.71 ^a	0.16	1.28, 2.14
<i>Aerococcus viridans</i>	1.45 ^{ab}	0.27	0.70, 2.2
<i>Corynebacterium bovis</i>	2.17 ^{ab}	0.37	1.14, 3.20
<i>Streptococcus dysgalactiae</i>	7.41 ^c	0.63	5.67, 9.16
Other IMI ²	2.11 ^b	0.17	1.63, 2.59
Milk yield as outcome			
IMI			
No growth (healthy)	8.95 ^b	0.45	7.69, 10.21
<i>Aerococcus viridans</i>	9.00 ^b	0.46	7.72, 10.28
<i>Corynebacterium bovis</i>	8.85 ^b	0.48	7.51, 10.19
<i>Streptococcus dysgalactiae</i>	7.62 ^a	0.58	6.00, 9.24
Other IMI ²	8.91 ^b	0.46	6.30, 8.70
Lactose as outcome ¹			
IMI			
No growth (healthy)	4.57 ^b	0.05	4.45, 4.70
<i>Aerococcus viridans</i>	4.62 ^b	0.06	4.46, 4.79
<i>Corynebacterium bovis</i>	4.59 ^{ab}	0.07	4.39, 4.79
<i>Streptococcus dysgalactiae</i>	4.38 ^{ab}	0.12	4.03, 4.73
Other IMI ²	4.51 ^a	0.05	4.38, 4.64
Protein as outcome ³			
IMI			
No growth (healthy)	3.20 ^a	0.08	2.99, 3.41
<i>Aerococcus viridans</i>	3.19 ^a	0.08	2.97, 3.41
<i>Corynebacterium bovis</i>	3.19 ^a	0.09	2.94, 3.44
<i>Streptococcus dysgalactiae</i>	3.39 ^a	0.11	3.08, 3.70
Other IMI ²	3.22 ^a	0.08	3.01, 3.43
Fat as outcome ³			
IMI			
No growth (healthy)	4.63 ^a	0.16	4.19, 5.08
<i>Aerococcus viridans</i>	4.27 ^a	0.28	3.50, 5.04
<i>Corynebacterium bovis</i>	4.27 ^a	0.46	3.01, 5.53
<i>Streptococcus dysgalactiae</i>	3.68 ^a	0.59	2.03, 5.32
Other IMI ²	4.48 ^a	0.18	4.19, 5.08

^{a-c}Estimates sharing the same superscript are not significantly different at the 5% level (Tukey-adjusted for multiple comparisons).

¹Estimates were derived when DIM was set to 150 to facilitate comparisons across different IMI statuses under standardized conditions.

²Other IMI includes IMI caused by *Staphylococcus* sp. or IMI caused by microorganisms that were isolated fewer than 30 times throughout the lactation period.

³Not statistically associated with the IMI status.

healthy quarters, but the magnitude of this increase was lower than that observed for IMI caused by *S. dysgalactiae* and *S. uberis*, and it was not significant after being adjusted for multiple comparisons. Similar findings have been reported in recent studies, where IMI caused by major pathogens were associated with a greater increase in SCC compared with both healthy quarters and those infected with minor pathogens (Schwarz et al., 2019, 2020b; Kirkeby et al., 2020).

The relationship between SCC and milk yield is well established, with elevated SCC consistently associated with reductions in milk production (Koldewij et al., 1999; Hagnestam-Nielsen et al., 2009; Hand et al., 2012; Huang et al., 2023). In our study, results for primiparous

cows were consistent with these findings, as quarters infected with *S. dysgalactiae* exhibited elevated SCS and a reduction of 1.3 kg/d in milk yield compared with healthy quarters. For multiparous cows, although IMI caused by *S. dysgalactiae* were associated with a substantial increase in SCS, no significant association with quarter milk yield was observed. Nonetheless, infected quarters demonstrated a numerical decrease in milk production compared with healthy quarters. Our results align with other studies that previously investigated the association of IMI caused by *S. dysgalactiae* and milk production. For example, a study conducted at the cow level reported a milk yield loss of 1.11 kg per test day for primiparous cows and 0.82 kg for multiparous cows infected with *S. dysgalactiae*, compared with cows negative for this pathogen (Whist et al., 2007). However, a recent study reported higher SCC but no difference in milk yield for cows subclinically infected with *S. dysgalactiae* compared with healthy cows (Zalewska et al., 2025). The difference between our findings regarding *S. dysgalactiae* IMI and those of this study might be partially attributed to the fact that we worked with quarter milk samples, which provide a more precise evaluation of udder health by avoiding the dilution effect (Reyher and Dohoo, 2011; Dufour and Dohoo, 2012).

Another pathogen commonly associated with a substantial increase in SCC is *S. uberis* (Woudstra et al., 2023). In our study, we found that IMI caused by *S. uberis* caused a substantial increase in SCS compared with healthy quarters; however, no difference, even numerical, was observed in milk production. A recent study using composite milk samples investigated the impact of subclinical mastitis on milk yield. This study reported higher SCC for cows infected with *S. uberis* compared with healthy cows; however, no difference in milk yield was observed, aligning with our findings (Zalewska et al., 2025). Moreover, we hypothesize that the lack of association between IMI caused by *S. uberis* and milk production may be partially attributed to the use of milk samples from apparently healthy quarters (i.e., subclinical mastitis). A study involving experimentally induced *S. uberis* infection using a strain isolated from a clinical mastitis case reported reduced milk yield in challenged cows compared with controls; however, this reduction was observed only between 3 and 5 d postchallenge (Kester et al., 2015).

Regarding milk components, we found that quarters infected with *S. dysgalactiae* for either primiparous or multiparous cows had a 0.2% reduction in lactose compared with healthy quarters. However, this decrease is likely too minor to hold any clinical relevance. No significant associations were found between IMI status and either protein or fat content in primiparous cows. However, a numerical difference of ~1% in fat content was observed

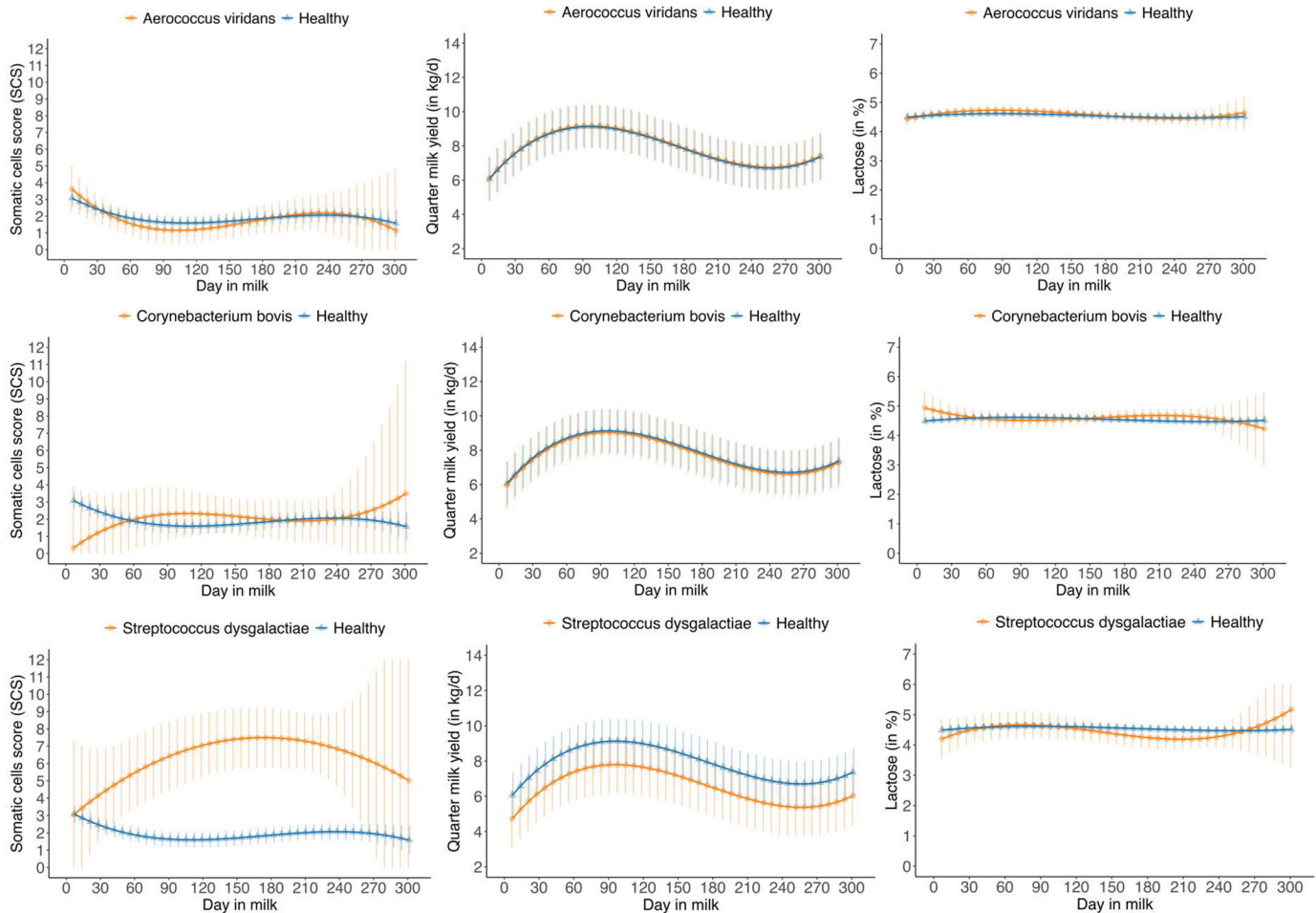


Figure 3. Estimates derived from the final mixed effects models for explaining the variation in SCS, milk yield (kg/d), and lactose during the lactation period according to the IMI status of primiparous cows and as a function of DIM. The liberal case definition was used in these models. Error bars represent the 95% CI.

for quarters infected by *S. dysgalactiae* compared with healthy quarters. Supporting our findings, Whist et al. (2007) reported no significant differences in lactose, protein, and fat content between *S. dysgalactiae*-positive and culture-negative cows. In multiparous cows, quarters infected with *S. uberis* showed a reduction of 1.0% in fat content compared with healthy quarters, and these differences might be relevant if many cows are infected. Similarly, Zalewska et al. (2025) found no association between IMI status (including *S. uberis*) and lactose, protein, or fat content. Furthermore, Kester et al. (2015) reported that cows experimentally infected with *S. uberis* had a small reduction in lactose (0.1%) but showed no differences in protein or fat content compared with the control cows.

In our study, quarters infected with minor pathogens, including *A. viridans*, *C. bovis*, *C. amycolatum*, and *L. garvieae*, showed no statistical difference in mean SCS

compared with healthy quarters. Additionally, no significant effect on milk yield or milk components was observed for these pathogens, except for *C. bovis*. For quarters infected with *C. bovis*, we observed a 0.8% reduction in fat content compared with healthy quarters. This difference might be relevant if a large number of cows are infected. Similar findings have been reported in the literature. For instance, Gonçalves et al. (2016) found that quarters infected with *C. bovis* showed increased SCC without a corresponding reduction in milk yield, as well as no effects on fat or protein content. Additionally, Gonçalves et al. (2020) and Zalewska et al. (2025) reported no association between infections caused by minor pathogens with milk yield or milk components, further supporting our findings, except for fat content.

Our findings should be interpreted in the context of emerging evidence from microbiome studies suggesting that microorganisms can be recovered from apparently

Table 4. Marginal estimates derived from the final mixed effects models for explaining the variation in the quarter SCS, milk yield (kg/d), and lactose (%) during the lactation period according to the IMI status of multiparous cows

Item	Marginal estimates	SE	95% CI
SCS as outcome ¹			
IMI			
No growth (healthy)	1.86 ^a	0.09	1.61, 2.11
<i>Corynebacterium amycolatum</i>	2.49 ^a	0.43	1.32, 3.67
<i>Corynebacterium bovis</i>	2.28 ^a	0.24	1.61, 2.95
<i>Lactococcus garvieae</i>	3.68 ^{ab}	0.71	1.72, 5.64
<i>Streptococcus dysgalactiae</i>	7.31 ^c	0.31	6.45, 8.18
<i>Streptococcus uberis</i>	6.48 ^c	0.37	5.45, 7.50
Other IMI ²	5.05 ^b	0.14	4.69, 5.44
Milk yield as outcome ³			
IMI			
No growth (healthy)	10.19 ^a	0.37	9.17, 11.20
<i>Corynebacterium amycolatum</i>	10.22 ^a	0.55	8.70, 11.70
<i>Corynebacterium bovis</i>	10.26 ^a	0.43	9.08, 11.40
<i>Lactococcus garvieae</i>	10.17 ^a	0.86	7.78, 12.60
<i>Streptococcus dysgalactiae</i>	9.48 ^a	0.54	7.98, 11.00
<i>Streptococcus uberis</i>	10.83 ^a	0.67	8.98, 12.70
Other IMI ²	9.96 ^a	0.39	8.88, 11.00
Lactose as outcome ¹			
IMI			
No growth (healthy)	4.37 ^a	0.03	4.29, 4.45
<i>Corynebacterium amycolatum</i>	4.27 ^{abc}	0.06	4.09, 4.45
<i>Corynebacterium bovis</i>	4.35 ^{ab}	0.04	4.23, 4.47
<i>Lactococcus garvieae</i>	4.33 ^{abc}	0.09	4.08, 4.57
<i>Streptococcus dysgalactiae</i>	4.13 ^c	0.07	3.92, 4.33
<i>Streptococcus uberis</i>	4.14 ^{abc}	0.09	3.88, 4.40
Other IMI ²	4.27 ^{bc}	0.03	4.18, 4.35
Protein as outcome ³			
IMI			
No growth (healthy)	3.18 ^a	0.03	3.09, 3.27
<i>Corynebacterium amycolatum</i>	3.21 ^a	0.06	3.05, 3.37
<i>Corynebacterium bovis</i>	3.15 ^a	0.04	3.03, 3.26
<i>Lactococcus garvieae</i>	3.31 ^a	0.08	3.08, 3.54
<i>Streptococcus dysgalactiae</i>	3.24 ^a	0.05	3.09, 3.39
<i>Streptococcus uberis</i>	3.20 ^a	0.06	3.03, 3.37
Other IMI ²	3.21 ^a	0.03	3.12, 3.31
Fat as outcome			
IMI			
No growth (healthy)	5.09 ^a	0.13	4.73, 5.46
<i>Corynebacterium amycolatum</i>	4.97 ^{ab}	0.40	3.87, 6.07
<i>Corynebacterium bovis</i>	4.30 ^b	0.26	3.58, 5.03
<i>Lactococcus garvieae</i>	4.86 ^{ab}	0.56	3.30, 6.41
<i>Streptococcus dysgalactiae</i>	4.59 ^{ab}	0.30	3.75, 5.43
<i>Streptococcus uberis</i>	4.05 ^b	0.35	3.07, 5.03
Other IMI ²	4.84 ^{ab}	0.16	4.41, 5.27

^{a-c}Estimates sharing the same superscript are not significantly different at the 5% level (Tukey-adjusted for multiple comparisons).

¹Estimates were derived when DIM was set to 150 to facilitate comparisons across different IMI statuses under standardized conditions.

²Other IMI includes IMI caused by *Staphylococcus* sp. or IMI caused by microorganisms that were isolated fewer than 30 times throughout the lactation period.

³Not statistically associated with the IMI status.

healthy mammary glands, although there is currently no consensus on whether the mammary gland is a nonsterile environment (Derakhshani et al., 2018; Pirard et al., 2021; Ruegg, 2022b). For several minor microorganisms included in this study, isolation of a bacterial species from

a quarter was not consistently associated with increased SCC or reduced milk yield, suggesting that their presence does not invariably elicit a measurable inflammatory or production response. This raises the possibility that some microorganisms may act as colonizers or have context-dependent pathogenicity rather than behaving as obligate mastitis pathogens. From a practical perspective, these results suggest that detection of certain microorganisms by culture alone may not necessarily indicate clinically or economically relevant IMI requiring intervention. Instead, their effect may depend on host factors, microbial interactions, or environmental conditions.

As with any study, certain limitations must be acknowledged. First, we used a convenience sample of dairy herds, which is not the optimal sampling strategy for accurately estimating the prevalence of IMI. Consequently, caution is needed when generalizing our prevalence findings to herds beyond those included in this study. Additionally, for our strict case definition, we used 2 samples taken 2 wk apart to define the IMI status. This approach has the disadvantage that infections must persist for at least 2 wk to be counted, thereby excluding shorter-duration infections. To overcome this, we have also provided the same analysis using the liberal case definition (Supplemental Material S8). Moreover, an important limitation in defining persistent infections was the lack of strain typing, which may have constrained our ability to accurately differentiate persistent infections from reinfections caused by different strains. Another limitation was the lack of history of mastitis treatment during the study period. Antimicrobial treatment can suppress bacterial growth or reduce inflammation, leading to lower SCC levels in the affected quarter. Furthermore, treatment might affect milk yield by mitigating the effect of infection on milk production. For instance, an animal recovering from mastitis treatment could show milk yield levels closer to healthy quarters despite underlying damage or stress caused by the prior infection. However, we believe the impact in our study is likely minimal, as the incidence rate of microorganisms that typically require treatment (e.g., *S. dysgalactiae* and *S. uberis*) was low. In addition, milk culture results were not communicated to producers until the end of the study and therefore could not have influenced their treatment practices during the study period. Lastly, we relied on bacteriological culture as the reference standard for classifying IMI, using a threshold of 1 cfu/10 µL to define infection. This approach resulted in ~3.6% and 5.4% of samples being categorized as contaminated for primiparous and multiparous datasets, respectively. Although this low threshold improves sensitivity, it may also have led to false positives, introducing potential misclassification bias, especially if bacteria from teat skin contaminated the milk samples.

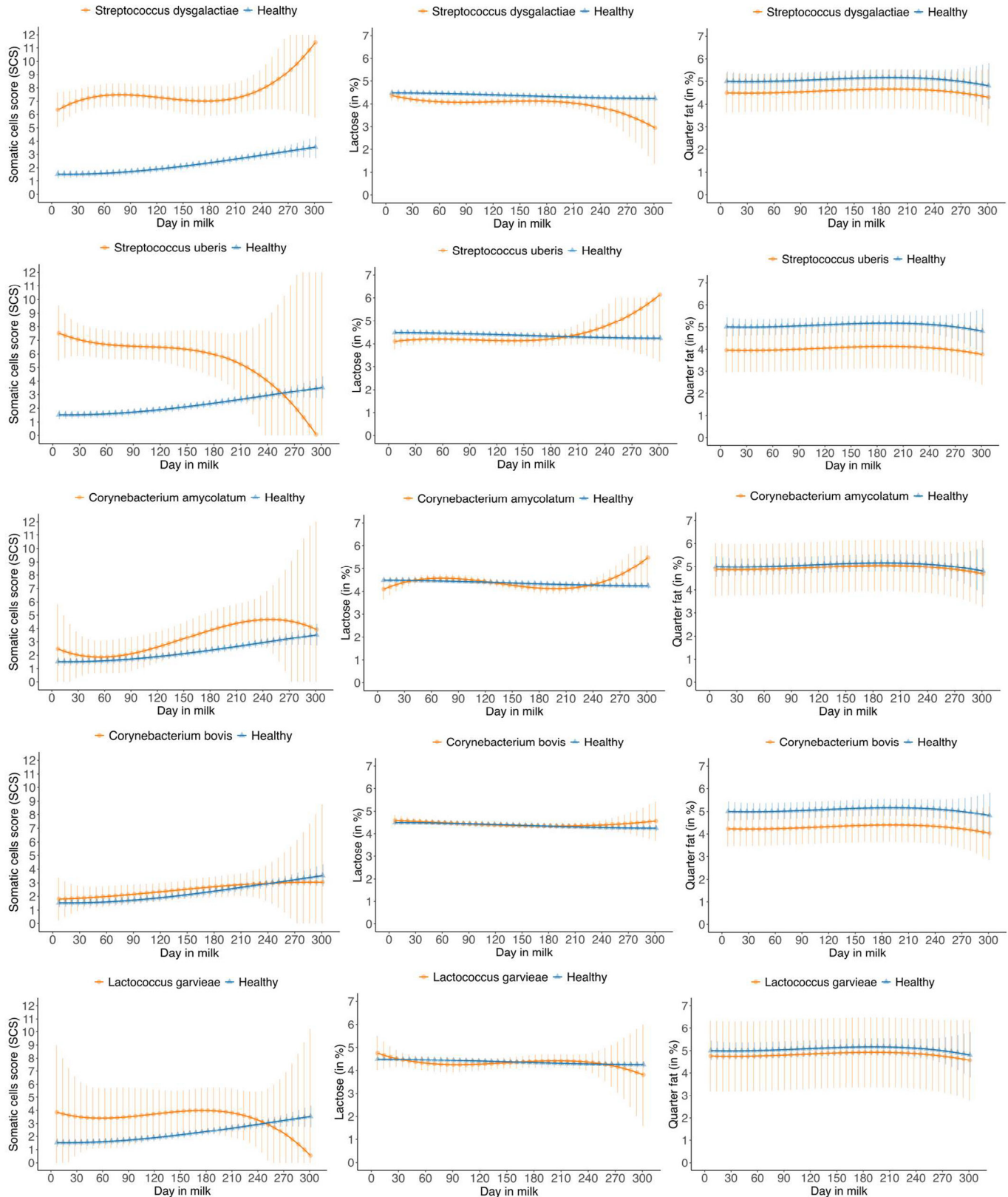


Figure 4. Estimates derived from the final mixed effects models for explaining the variation in SCS, lactose (%), and fat (%) during the lactation period according to the IMI status of multiparous cows and as a function of DIM. The conservative case definition was used in these models. Error bars represent the 95% CI.

CONCLUSIONS

Intramammary infections caused by *S. dysgalactiae* and *S. uberis* had a significant impact on SCS. In addition, IMI caused by *S. dysgalactiae* had a significant impact on milk yield, particularly in primiparous cows, whereas multiparous cows exhibited more subtle, numerical losses. The minimal effects on milk composition, except for fat content in quarters infected by *S. uberis*, *S. dysgalactiae*, and *C. bovis*, suggest that the primary economic burden lies in reduced milk production. In contrast, no significant differences on milk yield or milk composition were observed for the other studied microorganisms. The low incidence rates but prolonged persistence of infections caused by *S. dysgalactiae* and *S. uberis* highlight the need for targeted prevention and control strategies. Addressing these infections could mitigate their economic impact and improve overall herd health and productivity.

NOTES

This work was carried out with the financial support of the Agri-Food Innov'Action Program (Project Number IA 120635), a program resulting from the Growing Forward 2 agreement between the Québec Department of Agriculture, Fisheries and Food (MAPAQ, Québec, QC, Canada) and Agriculture and Agri-Food Canada (AAFC, Ottawa, ON, Canada). Moreover, the acquisition of the samples and data was made possible by funding from the Dairy Research Cluster 3 (Dairy Farmers of Canada, Ottawa, ON, Canada) and Agriculture and Agri-Food Canada (Ottawa, ON, Canada) under the Canadian Agricultural Partnership AgriScience Program and by the Mastitis Network (Saint-Hyacinthe, QC, Canada). The authors acknowledge the producers and the laboratory technicians who participated in the data collection. Supplemental material for this article is available at <https://doi.org/10.5683/SP3/TDOBTR>. Author contributions: MF performed the statistical analysis and prepared the first draft of the manuscript with tables and figures. SD contributed to the conceptual development, analysis, and writing of the manuscript. DK contributed to the data collection and curation. The manuscript was critically revised by JPR, DK, and SD. All authors contributed to the article and approved the submitted version. The project was approved by the Animal Care and Use Committee of the Faculty of Veterinary Medicine (protocol 18-Rech-1975). The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: AMS = automated milking system; IQR = interquartile range.

REFERENCES

- Aghamohammadi, M., D. Haine, D. F. Kelton, H. W. Barkema, H. Hogeveen, G. P. Keefe, and S. Dufour. 2018. Herd-level mastitis-associated costs on Canadian dairy farms. *Front. Vet. Sci.* 5:100. <https://doi.org/10.3389/fvets.2018.00100>.
- Bisutti, V., A. Vanzin, A. Toscano, S. Pegolo, D. Giannuzzi, F. Tagliapietra, S. Schiavon, L. Gallo, E. Trevisi, R. Negrini, and A. Cecchinato. 2022. Impact of somatic cell count combined with differential somatic cell count on milk protein fractions in Holstein cattle. *J. Dairy Sci.* 105:6447–6459. <https://doi.org/10.3168/jds.2022-22071>.
- Bobbo, T., C. Cipolat-Gotet, G. Bittante, and A. Cecchinato. 2016. The nonlinear effect of somatic cell count on milk composition, coagulation properties, curd firmness modeling, cheese yield, and curd nutrient recovery. *J. Dairy Sci.* 99:5104–5119. <https://doi.org/10.3168/jds.2015-10512>.
- Costa, A., N. Lopez-Villalobos, N. W. Sneddon, L. Shalloo, M. Franzoi, M. De Marchi, and M. Penasa. 2019. Invited review: Milk lactose—Current status and future challenges in dairy cattle. *J. Dairy Sci.* 102:5883–5898. <https://doi.org/10.3168/jds.2018-15955>.
- Davies, P. L., J. A. Leigh, A. J. Bradley, S. C. Archer, R. D. Emes, and M. J. Green. 2016. Molecular epidemiology of *Streptococcus uberis* clinical mastitis in dairy herds: Strain heterogeneity and transmission. *J. Clin. Microbiol.* 54:68–74. <https://doi.org/10.1128/JCM.01583-15>.
- Derakhshani, H., K. B. Fehr, S. Sepehri, D. Francoz, J. De Buck, H. W. Barkema, J. C. Plaizier, and E. Khafipour. 2018. Invited review: Microbiota of the bovine udder: Contributing factors and potential implications for udder health and mastitis susceptibility. *J. Dairy Sci.* 101:10605–10625. <https://doi.org/10.3168/jds.2018-14860>.
- Dufour, S., and I. R. Dohoo. 2012. Monitoring dry period intramammary infection incidence and elimination rates using somatic cell count measurements. *J. Dairy Sci.* 95:7173–7185. <https://doi.org/10.3168/jds.2012-5839>.
- Fonseca, M., D. Kurban, J.-P. Roy, D. E. Santschi, E. Molgat, and S. Dufour. 2025. Usefulness of differential somatic cell count for udder health monitoring: Association of differential somatic cell count and somatic cell score with quarter-level milk yield and milk components. *J. Dairy Sci.* 108:3900–3916. <https://doi.org/10.3168/jds.2024-25402>.
- Franzoi, M., C. L. Manuelian, M. Penasa, and M. De Marchi. 2020. Effects of somatic cell score on milk yield and mid-infrared predicted composition and technological traits of Brown Swiss, Holstein Friesian, and Simmental cattle breeds. *J. Dairy Sci.* 103:791–804. <https://doi.org/10.3168/jds.2019-16916>.
- Gonçalves, J. L., C. Kamphuis, H. Vernooij, J. P. Araújo Jr., R. C. Grenfell, L. Juliano, K. L. Anderson, H. Hogeveen, and M. V. Dos Santos. 2020. Pathogen effects on milk yield and composition in chronic subclinical mastitis in dairy cows. *Vet. J.* 262:105473. <https://doi.org/10.1016/j.tvjl.2020.105473>.
- Gonçalves, J. L., T. Tomazi, J. R. Barreiro, D. C. Beuron, M. A. Arcari, S. H. I. Lee, C. M. de Magalhães Rodrigues Martins, J. P. Araújo Junior, and M. V. dos Santos. 2016. Effects of bovine subclinical mastitis caused by *Corynebacterium* spp. on somatic cell count, milk yield and composition by comparing contralateral quarters. *Vet. J.* 209:87–92. <https://doi.org/10.1016/j.tvjl.2015.08.009>.
- Hagnestam-Nielsen, C., U. Emanuelson, B. Berglund, and E. Strandberg. 2009. Relationship between somatic cell count and milk yield in different stages of lactation. *J. Dairy Sci.* 92:3124–3133. <https://doi.org/10.3168/jds.2008-1719>.
- Hand, K. J., A. Godkin, and D. F. Kelton. 2012. Milk production and somatic cell counts: A cow-level analysis. *J. Dairy Sci.* 95:1358–1362. <https://doi.org/10.3168/jds.2011-4927>.
- Hogan, J. S., R. N. Gonzalez, R. J. Harmon, S. C. Nickerson, S. P. Oliver, J. W. Pankey, and K. L. Smith. 1999. Laboratory Handbook on Bovine Mastitis. Rev. ed. National Mastitis Council, Madison, WI.
- Huang, C.-H., K. Furukawa, and N. Kusaba. 2023. Estimating the nonlinear interaction between somatic cell score and differential somatic cell count on milk production by parity using generalized additive models. *J. Dairy Sci.* 106:7942–7953. <https://doi.org/10.3168/jds.2022-22958>.

- Kester, H. J., D. E. Sorter, and J. S. Hogan. 2015. Activity and milk compositional changes following experimentally induced *Streptococcus uberis* bovine mastitis. *J. Dairy Sci.* 98:999–1004. <https://doi.org/10.3168/jds.2014-8576>.
- Kirkeby, C., N. Toft, D. Schwarz, M. Farre, S. S. Nielsen, L. Zervens, S. Hechinger, and T. Halasa. 2020. Differential somatic cell count as an additional indicator for intramammary infections in dairy cows. *J. Dairy Sci.* 103:1759–1775. <https://doi.org/10.3168/jds.2019-16523>.
- Koldeweij, E., U. Emanuelson, and L. Janson. 1999. Relation of milk production loss to milk somatic cell count. *Acta Vet. Scand.* 40:47–56. <https://doi.org/10.1186/BF03547040>.
- Kurban, D., J.-P. Roy, F. Kabera, A. Fréchette, M. M. Um, A. Albaaj, S. Rowe, S. Godden, P. R. F. Adkins, J. R. Middleton, M.-L. Gauthier, G. P. Keefe, T. J. DeVries, D. F. Kelton, P. Moroni, M. Veiga Dos Santos, H. W. Barkema, and S. Dufour. 2022. Diagnosing intramammary infection: Meta-analysis and mapping review on frequency and udder health relevance of microorganism species isolated from bovine milk samples. *Animals (Basel)* 12:3288. <https://doi.org/10.3390/ani12233288>.
- Kurban, D., J.-P. Roy, T. J. DeVries, G. P. Keefe, P. R. F. Adkins, J. R. Middleton, and S. Dufour. 2026. Dynamics of intramammary infections and their association with quarter-level somatic cell score, milk yield, and milk components. Part 1: *Staphylococcal* species. *J. Dairy Sci.* 109:7474–7494. <https://doi.org/10.3168/jds.2025-27763>.
- Olde Riekerink, R. G. M., H. W. Barkema, D. F. Kelton, and D. T. Scholl. 2008. Incidence rate of clinical mastitis on Canadian dairy farms. *J. Dairy Sci.* 91:1366–1377. <https://doi.org/10.3168/jds.2007-0757>.
- Pegolo, S., D. Giannuzzi, V. Bisutti, R. Tessari, M. E. Gelain, L. Gallo, S. Schiavon, F. Tagliapietra, E. Trevisi, P. Ajmone Marsan, G. Bittante, and A. Cecchinato. 2021. Associations between differential somatic cell count and milk yield, quality, and technological characteristics in Holstein cows. *J. Dairy Sci.* 104:4822–4836. <https://doi.org/10.3168/jds.2020-19084>.
- Penry, J. F., P. M. Crump, L. L. Hernandez, and D. J. Reinemann. 2018. Association of quarter milking measurements and cow-level factors in an automatic milking system. *J. Dairy Sci.* 101:7551–7562. <https://doi.org/10.3168/jds.2017-14153>.
- Piepers, S., L. De Meulemeester, A. De Kruif, G. Opsomer, H. W. Barkema, and S. De Vliegher. 2007. Prevalence and distribution of mastitis pathogens in subclinically infected dairy cows in Flanders, Belgium. *J. Dairy Res.* 74:478–483. <https://doi.org/10.1017/S0022029907002841>.
- Pirard, B., S. Crèvecoeur, P. A. Fall, P. Lausberg, B. Taminiau, and G. Daube. 2021. Potential resident bacterial microbiota in udder tissues of culled cows sampled in abattoir. *Res. Vet. Sci.* 136:369–372. <https://doi.org/10.1016/j.rvsc.2021.03.012>.
- Reyher, K. K., and I. R. Dohoo. 2011. Diagnosing intramammary infections: Evaluation of composite milk samples to detect intramammary infections. *J. Dairy Sci.* 94:3387–3396. <https://doi.org/10.3168/jds.2010-3907>.
- Ruegg, P. L. 2022a. Realities, challenges and benefits of antimicrobial stewardship in dairy practice in the United States. *Microorganisms* 10:1626. <https://doi.org/10.3390/microorganisms10081626>.
- Ruegg, P. L. 2022b. The bovine milk microbiome—An evolving science. *Domest. Anim. Endocrinol.* 79:106708. <https://doi.org/10.1016/j.domaniend.2021.106708>.
- Sargeant, J. M., A. M. O'Connor, I. R. Dohoo, H. N. Erb, M. Cevallos, M. Egger, A. K. Ersbøll, S. W. Martin, L. R. Nielsen, D. L. Pearl, D. U. Pfeiffer, J. Sanchez, M. E. Torrence, H. Vigre, C. Waldner, and M. P. Ward. 2016. Methods and processes of developing the strengthening the reporting of observational studies in epidemiology – veterinary (STROBE-Vet) statement. *Prev. Vet. Med.* 134:188–196. <https://doi.org/10.1016/j.prevetmed.2016.09.005>.
- Schukken, Y. H., J. Günther, J. Fitzpatrick, M. C. Fontaine, L. Goetze, O. Holst, J. Leigh, W. Petzl, H.-J. Schuberth, A. Sipka, D. G. E. Smith, R. Quesnell, J. Watts, R. Yancey, H. Zerbe, A. Gurjar, R. N. Zadoks, and H.-M. Seyfert. 2011. Host-response patterns of intramammary infections in dairy cows. *Vet. Immunol. Immunopathol.* 144:270–289. <https://doi.org/10.1016/j.vetimm.2011.08.022>.
- Schwarz, D., S. Kleinhans, G. Reimann, P. Stückler, F. Reith, K. Ilves, K. Pedastsaar, L. Yan, Z. Zhang, M. Valdivieso, M. L. Barreal, and R. Fouz. 2020a. Investigation of dairy cow performance in different udder health groups defined based on a combination of somatic cell count and differential somatic cell count. *Prev. Vet. Med.* 183:105123. <https://doi.org/10.1016/j.prevetmed.2020.105123>.
- Schwarz, D., Z. Lipkens, S. Piepers, and S. De Vliegher. 2019. Investigation of differential somatic cell count as a potential new supplementary indicator to somatic cell count for identification of intramammary infection in dairy cows at the end of the lactation period. *Prev. Vet. Med.* 172:104803. <https://doi.org/10.1016/j.prevetmed.2019.104803>.
- Schwarz, D., D. E. Santschi, J. Durocher, and D. M. Lefebvre. 2020b. Evaluation of the new differential somatic cell count parameter as a rapid and inexpensive supplementary tool for udder health management through regular milk recording. *Prev. Vet. Med.* 181:105079. <https://doi.org/10.1016/j.prevetmed.2020.105079>.
- Sharma, N., N. K. Singh, and M. S. Bhadwal. 2011. Relationship of somatic cell count and mastitis: An overview. *Asian-Australas. J. Anim. Sci.* 24:429–438. <https://doi.org/10.5713/ajas.2011.10233>.
- Shook, G. E. 1993. Genetic improvement of mastitis through selection on somatic cell count. *Vet. Clin. North Am. Food Anim. Pract.* 9:563–577. [https://doi.org/10.1016/S0749-0720\(15\)30622-8](https://doi.org/10.1016/S0749-0720(15)30622-8).
- Silva Boloña, P., J. Upton, V. Cabrera, T. Erker, and D. J. Reinemann. 2022. Simulation model of quarter milk flowrates to estimate quarter and cow milking duration and automated milking system's box duration. *J. Dairy Sci.* 105:4156–4170. <https://doi.org/10.3168/jds.2021-20464>.
- Vakkamäki, J., S. Taponen, A.-M. Heikkilä, and S. Pyörälä. 2017. Bacteriological etiology and treatment of mastitis in Finnish dairy herds. *Acta Vet. Scand.* 59:33. <https://doi.org/10.1186/s13028-017-0301-4>.
- von Keyserlingk, M. A. G., J. Rushen, A. M. De Passillé, and D. M. Weary. 2009. Invited review: The welfare of dairy cattle—Key concepts and the role of science. *J. Dairy Sci.* 92:4101–4111. <https://doi.org/10.3168/jds.2009-2326>.
- Wever, P., and U. Emanuelson. 1989. Effects of systematic influences and intramammary infection on differential and total somatic cell counts in quarter milk samples from dairy cows. *Acta Vet. Scand.* 30:465–474. <https://doi.org/10.1186/BF03548024>.
- Whist, A. C., O. Østerås, and L. Sølvørød. 2007. *Streptococcus dysgalactiae* isolates at calving and lactation performance within the same lactation. *J. Dairy Sci.* 90:766–778. [https://doi.org/10.3168/jds.S0022-0302\(07\)71561-8](https://doi.org/10.3168/jds.S0022-0302(07)71561-8).
- Woudstra, S., N. Wente, Y. Zhang, S. Leimbach, M. K. Gussmann, C. Kirkeby, and V. Krömker. 2023. Strain diversity and infection durations of *Staphylococcus* spp. and *Streptococcus* spp. causing intramammary infections in dairy cows. *J. Dairy Sci.* 106:4214–4231. <https://doi.org/10.3168/jds.2022-22942>.
- Zalewska, M., P. Brzozowska, M. Rzewuska, E. Kawecka-Grochocka, D. M. Urbańska, T. Sakowski, and E. Bagnicka. 2025. The quality and technological parameters of milk obtained from dairy cows with subclinical mastitis. *J. Dairy Sci.* 108:1285–1300. <https://doi.org/10.3168/jds.2024-25346>.
- Zecconi, A., G. Meroni, V. Sora, R. Mattina, M. Cipolla, and L. Zanini. 2021. Total and differential cell counts as a tool to identify intramammary infections in cows after calving. *Animals (Basel)* 11:727. <https://doi.org/10.3390/ani11030727>.

ORCID

Mariana Fonseca, <https://orcid.org/0000-0002-6383-2547>
 Daryna Kurban, <https://orcid.org/0000-0002-9298-9076>
 Jean-Philippe Roy, <https://orcid.org/0000-0002-0444-2303>
 Simon Dufour <https://orcid.org/0000-0001-6418-0424>